# Matplotlib Cheat Sheet

```
import matplotlib.pyplot as plt
```

## Customize Plot

Set default values for fontsize and linewidth

```
plt.rcParams["font.size"] = "13"
plt.rcParams["lines.linewidth"] = "2"
```

**Title and axis labels**

```
plt.title("Heat Equation", fontweight="bold")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
```

**Legend**

```
plt.legend(loc="right",
        facecolor=(0.9, 0.9, 0.9))
```

```
ax = plt.gca()
ax.set_aspect("equal")    # Don't stretch the plot
plt.grid(b=True)          # enable grid
```

Move axis to origin

```
ax.spines["bottom"].set_position("zero")
ax.spines["left"].set_position("zero")
```

Box off

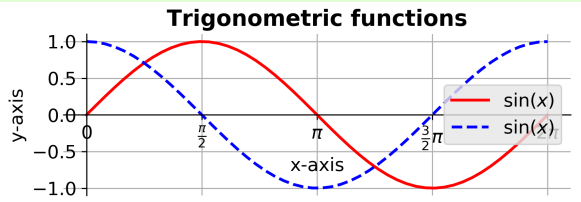
```
ax.spines["top"].set_color("none")
ax.spines["right"].set_color("none")
```

Set x- and y-ticks, as well as labels

```
plt.xticks([0, np.pi, 2*np.pi])
ax.set_xticklabels([r'$0$', r'$\pi$', r'$2\pi$'])
```

Save plot to a png-file, show result:

```
fig.set_size_inches(19.2, 10.8)
plt.savefig("plot_python.png", dpi=300)
plt.show()
```



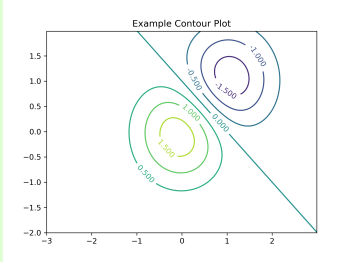

## Plot Types

**Simple plots:**

```
plt.plot(x, y, "xb-", label=r'$f(x)$' )
plt.scatter(x, y, c=None, marker=None)
```

**Mathematical plots:**

```
plt.contourf([X, Y], Z, [levels])
plt.quiver(X, Y, U, V, C)
plt.streamplot(X, Y, U, V, color=C)
ax.plot_surface(X, Y, Z)
ax.plot_wireframe(X, Y, Z)
```




Diagrams and Data Plots:

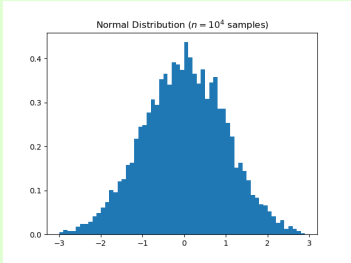
```
plt.bar(x, height, width=0.8)
plt.boxplot(data)
plt.errorbar(x, y, yerr=None, xerr=None)
```

Pie chart:

```
labels = np.array(["txtA", "txtB", "txtC"])
explode = 0.1 * np.array([1, 0, 1 ])
C = np.array(["cyan", "green", "yellow"])

plt.pie(x, explode=explode, labels=labels, c
```

Histogram:

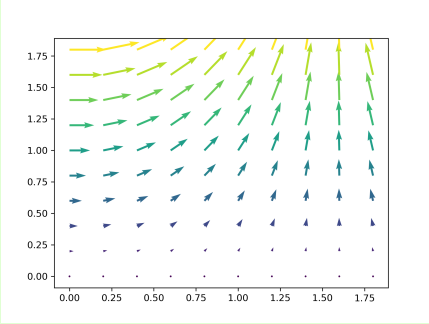
```
# Draw 10^4 random samples, mu=0, sigma=1
data = np.random.normal(0, 1, 10_000)
bins = np.arange(-3, 3, 0.1)

# Normalised histogram with density=True
plt.hist(data, bins, density=True)
```




## Quiver-plot

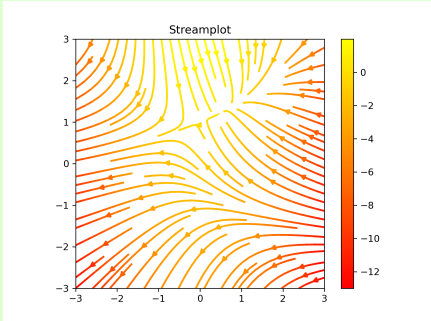
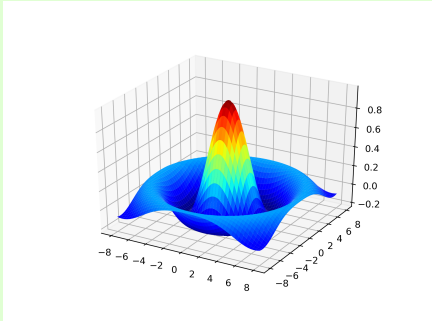
Plot vector fields via `plt.quiver()`:

```
# Create meshgrid
X = np.arange(0, 2, 0.2)
Y = X
X, Y = np.meshgrid(X, Y)

# Define function and color
U = np.cos(X) * Y
V = np.sin(X) * Y
C = np.sqrt(U**2 + V**2)

plt.quiver(X, Y, U, V, C,
                units="xy")
```



## Streamplot

```
Y, X = np.mgrid[-3:3:100j, -3:3:100j]
U = -1 - X**2 + Y
V = 1 + X - Y**2
speed = np.sqrt(U**2 + V**2)

strm = ax1.streamplot(X, Y, U, V, color=U, linewidth=2, cmap='autumn')
fig.colorbar(strm.lines)
```

## 3D-Surface

```
fig = plt.figure()
ax = fig.gca(projection='3d')

# Create data
X = Y = np.linspace(-8, 8, 100)
X, Y = np.meshgrid(X, Y)
R = np.sqrt(X**2 + Y**2) + 2e-16
Z = np.sin(R) / R

surf = ax.plot_surface(X, Y, Z, cmap="jet", linewidth=0)
```